Simulation Error-Choice Report

2026-05-21

1 Overview

This report evaluates multinomial sampling error across cell-type proportions generated from a weighted Beta($\alpha, 1$) distribution.

For each α, the workflow:

1. Generates true proportions over K cell types.
2. Simulates multinomial (no overdispersion) count draws of n cells, B times.
3. Computes maximum error across cell types for Absolute Error (AE) and Absolute Relative Error (ARE).
4. Summarizes threshold-based success rates and the cell-type index most frequently driving the maximum error.

2 Simulation setup

Table 1: Simulation configuration used in this report.

Parameter	Value
alpha	2, 2.5, 3, 4, 5
K	10
n	1000
B	5000
metrics	AE, ARE
model	multinomial
tie_method	random
seed	260926

3 True proportions from Beta(α, 1)

The 10 cell types are given proportions equal to the 0.05, 0.15, 0.25, ... 0.85, 0.95 quantiles from a Beta(α, 1) distribution and then weighted such that $\sum p_i = 1$.

Larger values of α tend to result in more small-proportion cell types.

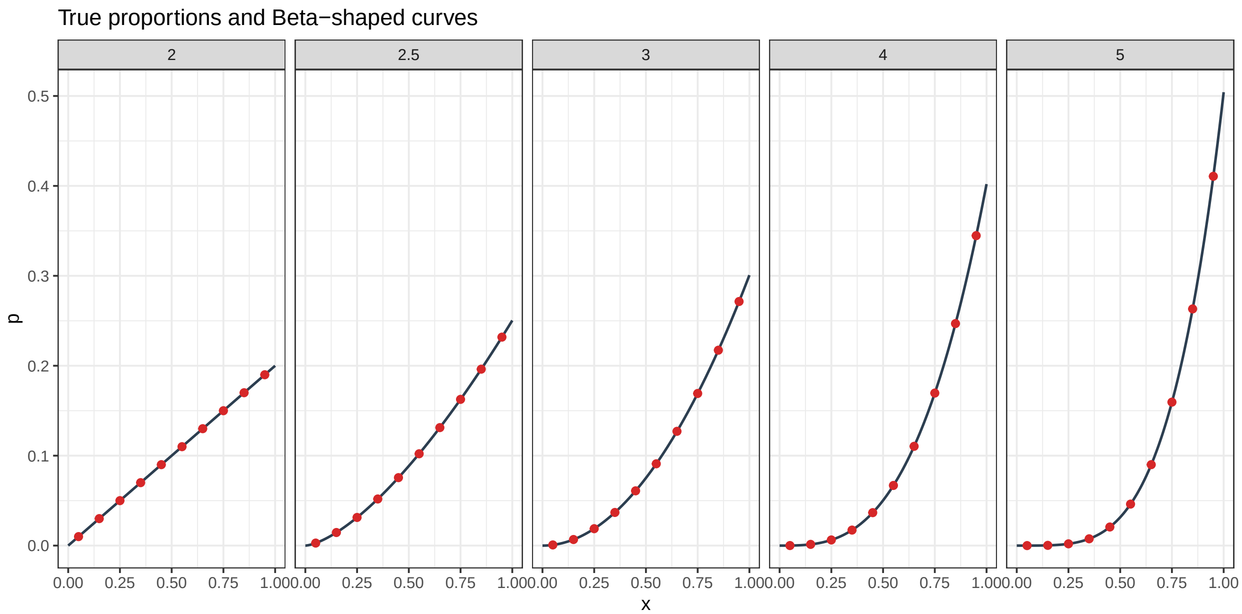

Figure 1: Weighted Beta(alpha,1) curves and grid-point proportions by alpha.

Table 2: True proportion values by alpha and cell-type index.

alpha	index_1	index_2	index_3	index_4	index_5	index_6	index_7	index_8	index_9	index_10
2.0	0.01000	0.03000	0.05000	0.07000	0.09000	0.11000	0.13000	0.15000	0.17000	0.19000
2.5	0.00280	0.01454	0.03129	0.05184	0.07558	0.10212	0.13120	0.16261	0.19620	0.23182
3.0	0.00075	0.00677	0.01880	0.03684	0.06090	0.09098	0.12707	0.16917	0.21729	0.27143
4.0	0.00005	0.00136	0.00628	0.01724	0.03663	0.06688	0.11040	0.16960	0.24688	0.34467
5.0	0.00000	0.00026	0.00197	0.00757	0.02068	0.04614	0.09000	0.15953	0.26319	0.41067

4 Success-rate curves

The success-rate curves show the probability of success (i.e. P(max(error)< threshold)) for different thresholds.

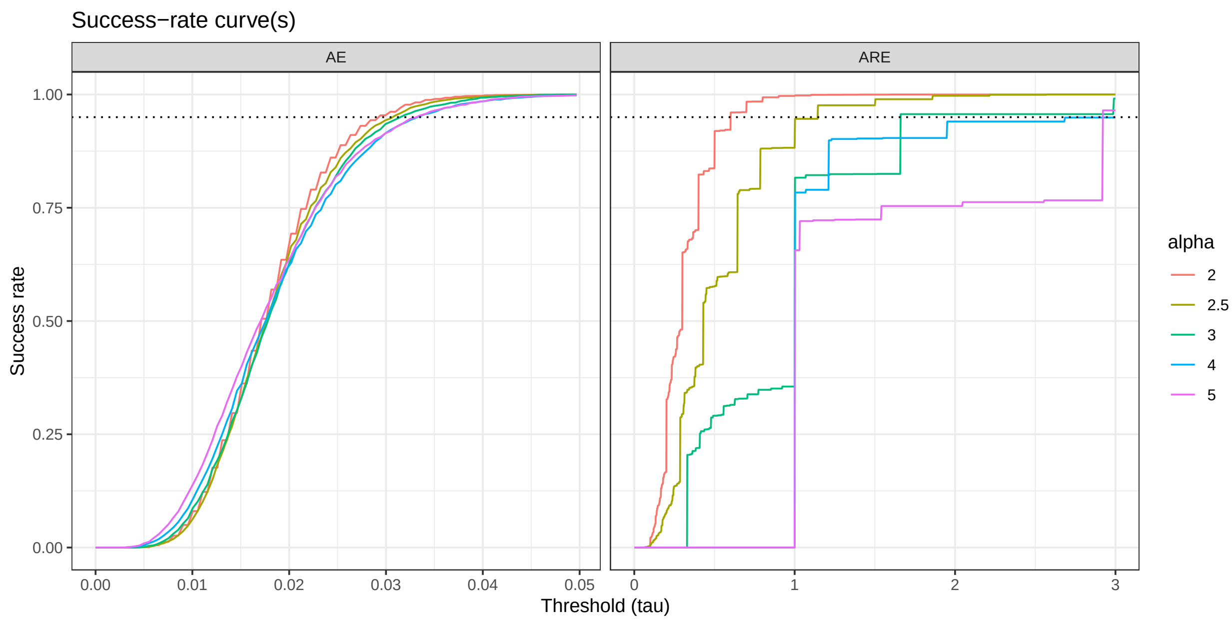

Figure 2: Success-rate curves by alpha, shown separately for AE and ARE.

For the AE, success rate curves are very similar for different true proportions vectors, with a 95% success rate for thresholds between 0.3 and 0.35.

For the ARE, success rate drops down for larger values of α. This is expected, as larger values of α increases the number of low-proportion cell types.

Note the jump in success rate for the ARE when threshold $\tau = 1$. Since ARE = AE / p, when ARE = 1, the AE is equal to the true proportion p. This happens when the estimated proportion $\hat{p}$ is equal to 0: i.e. the simulated cell count is 0.

5 Argmax histograms

The argmax histograms show the index at which the error is largest (larger indices represent larger true probabilities). In case of a tie, one index is chosen at random.

As expected, AE is largest for larger true proportions p (closer to 0.5). However, smaller proportions can still contribute to the largest AE.

The ARE is driven mostly by the 2-3 cell types with the smallest proportion. Note for $\alpha = 5$, the 2nd smallest proportion actually has the largest ARE, most of the time. The smallest proportion will often have a simulated cell count of 0, resulting in an ARE of exactly 1. The second smallest proportion then has a possibility to show a larger ARE.

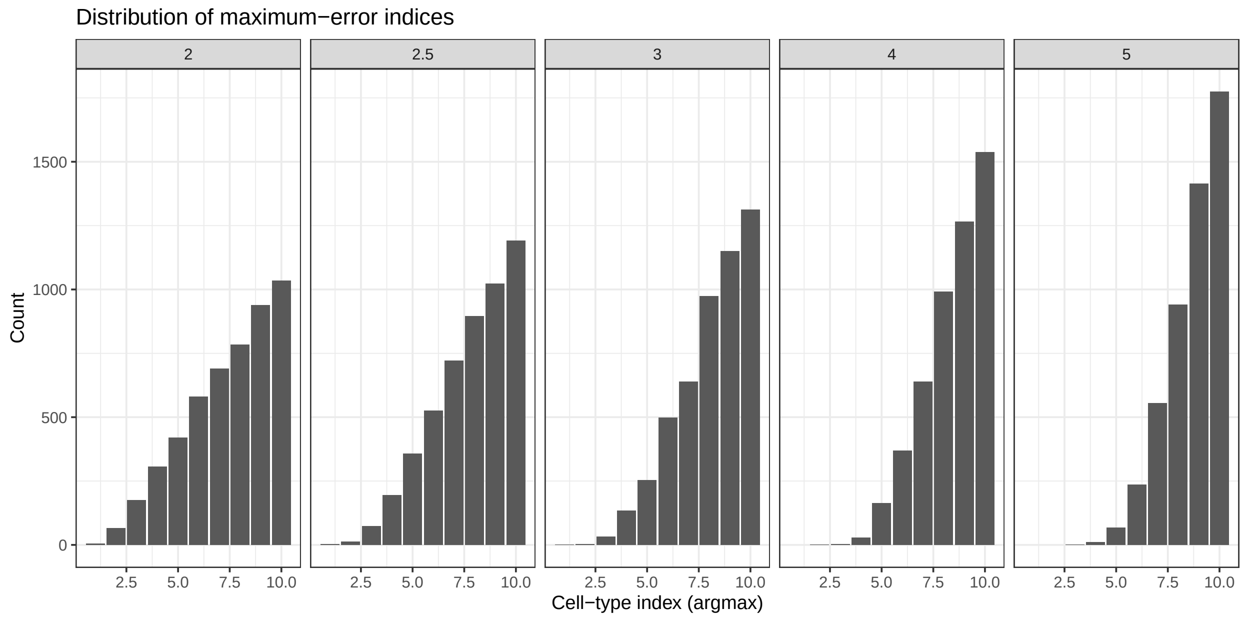

Figure 3: Argmax histograms by alpha for AE maximum-error indices.

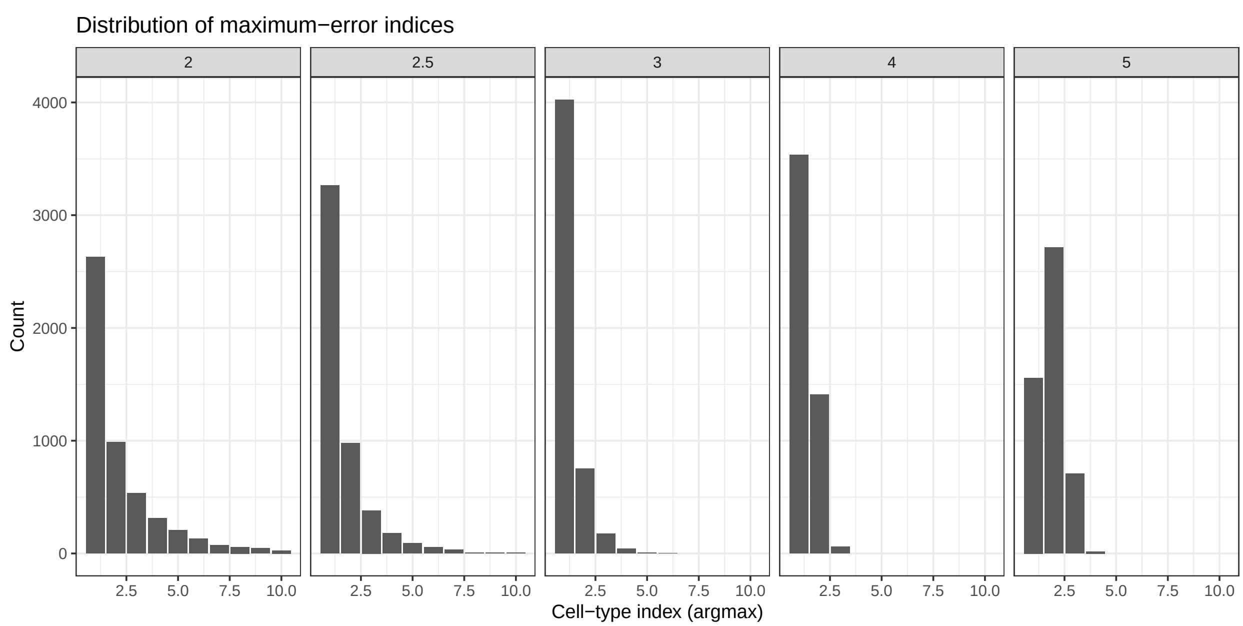

Figure 4: Argmax histograms by alpha for ARE maximum-error indices.

6 Keep highest proportion fixed, vary other proportions

6.1 Simulation proportions

The highest proportions are kept constant (0.2, 0.3, 0.4, 0.5, 0.6), with the other proportions varying based on a beta distribution (as before). Some values of alpha would lead to some of the remaining proportions to be larger than the maximal proportion and are thus excluded.

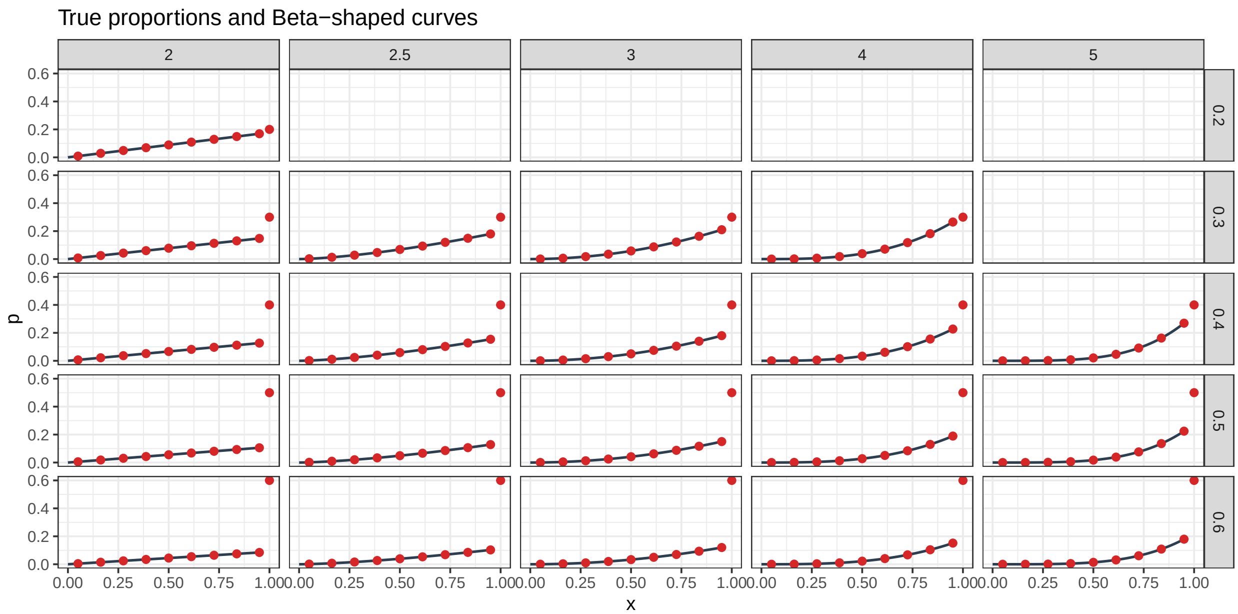

Figure 5: Grid-point proportions by alpha (columns), and maximal proportions (rows).

Table 3: True proportion values by alpha, maximal proportion and cell-type index.

alpha	p_max	index_1	index_2	index_3	index_4	index_5	index_6	index_7	index_8	index_9	index_10
2.0	0.2	0.00889	0.02889	0.04889	0.06889	0.08889	0.10889	0.12889	0.14889	0.16889	0.2
2.0	0.3	0.00778	0.02528	0.04278	0.06028	0.07778	0.09528	0.11278	0.13028	0.14778	0.3
2.5	0.3	0.00217	0.01272	0.02800	0.04684	0.06866	0.09309	0.11988	0.14883	0.17981	0.3
3.0	0.3	0.00058	0.00614	0.01759	0.03493	0.05815	0.08726	0.12226	0.16315	0.20993	0.3
4.0	0.3	0.00004	0.00133	0.00643	0.01799	0.03865	0.07104	0.11782	0.18162	0.26508	0.3
2.0	0.4	0.00667	0.02167	0.03667	0.05167	0.06667	0.08167	0.09667	0.11167	0.12667	0.4
2.5	0.4	0.00186	0.01090	0.02400	0.04015	0.05885	0.07979	0.10275	0.12757	0.15412	0.4
3.0	0.4	0.00050	0.00526	0.01508	0.02994	0.04984	0.07480	0.10480	0.13984	0.17994	0.4
4.0	0.4	0.00003	0.00114	0.00551	0.01542	0.03313	0.06089	0.10099	0.15567	0.22721	0.4
5.0	0.4	0.00000	0.00023	0.00189	0.00745	0.02066	0.04653	0.09133	0.16264	0.26926	0.4
2.0	0.5	0.00556	0.01806	0.03056	0.04306	0.05556	0.06806	0.08056	0.09306	0.10556	0.5
2.5	0.5	0.00155	0.00909	0.02000	0.03346	0.04904	0.06649	0.08563	0.10631	0.12844	0.5
3.0	0.5	0.00042	0.00439	0.01256	0.02495	0.04154	0.06233	0.08733	0.11654	0.14995	0.5
4.0	0.5	0.00003	0.00095	0.00459	0.01285	0.02761	0.05075	0.08416	0.12973	0.18934	0.5
5.0	0.5	0.00000	0.00019	0.00158	0.00621	0.01722	0.03877	0.07611	0.13553	0.22439	0.5
2.0	0.6	0.00444	0.01444	0.02444	0.03444	0.04444	0.05444	0.06444	0.07444	0.08444	0.6
2.5	0.6	0.00124	0.00727	0.01600	0.02677	0.03923	0.05319	0.06850	0.08505	0.10275	0.6
3.0	0.6	0.00033	0.00351	0.01005	0.01996	0.03323	0.04987	0.06987	0.09323	0.11996	0.6

alpha	p_max	index_1	index_2	index_3	index_4	index_5	index_6	index_7	index_8	index_9	index_10
4.0	0.6	0.00002	0.00076	0.00367	0.01028	0.02208	0.04060	0.06733	0.10378	0.15148	0.6
5.0	0.6	0.00000	0.00015	0.00126	0.00497	0.01377	0.03102	0.06089	0.10843	0.17951	0.6

6.2 Results

The highest proportion (closest to 0.5) has most often the largest AE. Additionally, for a given maximal proportion, the number of times this maximal proportion results in the largest AE seems to be stable over all values of alpha.

This could be repeated with a constant minimal proportion to check the effect on ARE.

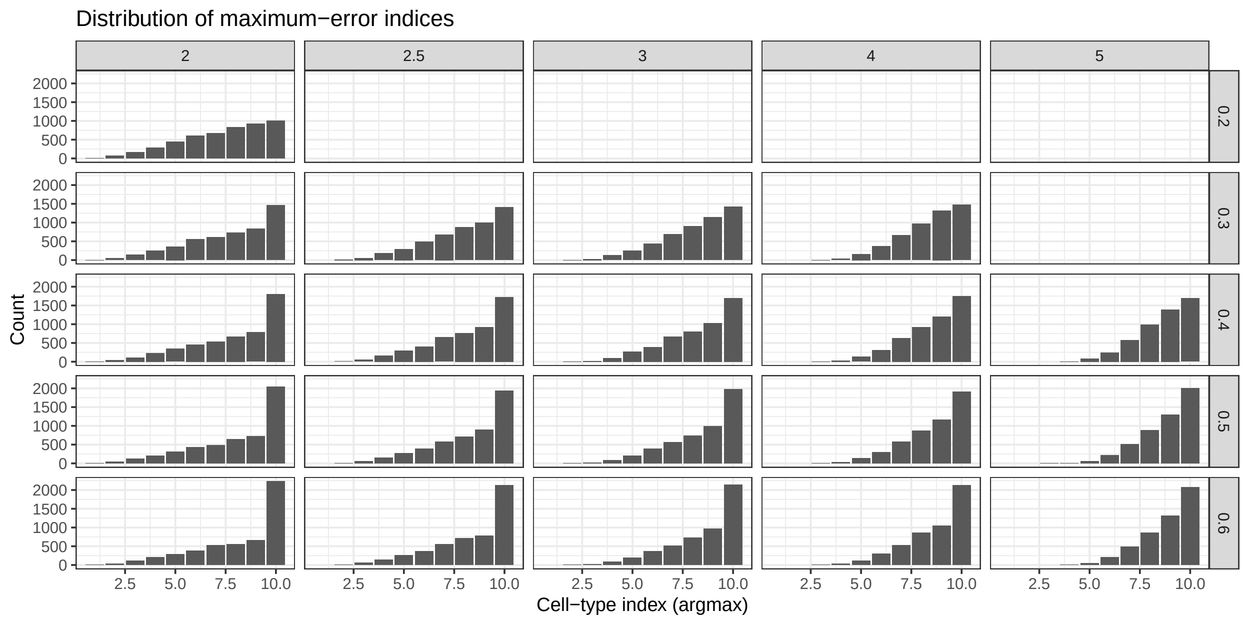

Figure 6: Argmax histograms by alpha (columns) and maximal proportion (rows) for AE maximum-error indices.

7 Next steps

- Repeat with overdispersion (dirichlet multinomial)
- Repeat with a fixed minimal proportion and check the effect on the ARE
- Find a cutoff for when to choose AE vs ARE
    - Idea: for a given threshold for AE and ARE, which cutoff for $\hat{p}$ maximizes the success-rate
- Hierarchical distributions
    - Idea: "parent" has a certain proportion,
    - "Daughter" cell types have (uniform) distribution
    - How does this affect the distribution of error metrics?